

Palladium/palladium Oxide/calcium Silicate/graphitic Carbon Nitride
Nanocomposite Material

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Abstract

A Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material includes spherical metal oxide nanoparticles, including a palladium@palladium oxide (Pd@PdO) phase and a calcium silicate (CaSiO₃) phase dispersed on a matrix of graphitic carbon nitride (g-C₃N₄) nanosheets. The spherical metal oxide nanoparticles have an average particle diameter in a range from 25 to 250 nanometers (nm). The Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a Brunauer-Emmett-Teller (BET) surface area of at least 45 square meters per gram (m²·g⁻¹). A method of preparing the nanocomposite material includes forming CaSiO₃, g-C₃N₄, and Pd@PdO intermediates and combining in a microwave reaction to form the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material.

Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure is directed to a nanocomposite material, and more particularly, to a nanocomposite material including palladium@palladium oxide (Pd@PdO), calcium silicate (CaSiO₃), and graphitic carbon nitride (g-C₃N₄) for use in nanotechnology applications.

Description of Related Art

(2) The “background” description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description which may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(3) Carbon-based nanomaterials (CNMs) such as graphene, carbon nanotubes (CNTs), and carbon nanofibers (CNFs) have garnered attention in nanotechnology due to their unique combination of high surface area, excellent electrical conductivity, mechanical strength, and chemical stability. These characteristics make them suitable for applications in energy storage systems like supercapacitors and lithium-ion batteries, where rapid charge transport and large active surface areas are useful. Additionally, their lightweight and flexible nature supports their integration into various devices, including flexible electronics and sensors. However, despite their outstanding properties, CNMs often show limited electrochemical activity and lack the required chemical functionalities for broader applications such as catalysis or ion exchange, which can restrict their overall performance when used alone.

(4) To overcome these limitations, multicomponent nanocomposites have been developed by integrating CNMs with electrochemically active materials such as metal oxides (e.g., MnO₂, TiO₂, Fe₂O₃) and metal silicates (e.g., CaSiO₃, MgSiO₃). These nanocomposites leverage the conductive framework of CNMs to facilitate rapid electron transport, while the metal-based components introduce enhanced redox activity, ion exchange capacity, and catalytic functionality. This synergy improves properties like specific capacitance, charge-discharge efficiency, structural stability, and long-term performance. Moreover, the carbon matrix helps to prevent the agglomeration of metal particles and enhances dispersion, providing uniform performance throughout the material. As a result, these hybrid nanocomposites offer advantages over single-component systems and are being actively explored in a wide range of nanotechnology applications, including energy storage, environmental remediation, catalysis, and sensor technologies.

(5) Although multicomponent nanocomposite materials have been developed in the past, there still exists a need to develop nanocomposites with improved properties. Accordingly, it is one object of

the present disclosure to provide a metal-metal oxide-metal silicate-carbon based quaternary nanocomposite by integrating metal, metal oxides, metal silicate, and carbon nanomaterials to enhance the properties and their synergistic effects, making them ideal for applications in energy storage, catalysis, biomedicine, and environmental sustainability.

SUMMARY

(6) In an exemplary embodiment, a Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material is described. The Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material includes spherical metal oxide nanoparticles, including a palladium@palladium oxide (Pd@PdO) phase and a calcium silicate (CaSiO₃) phase dispersed on a matrix of graphitic carbon nitride (g-C₃N₄) nanosheets. The spherical metal oxide nanoparticles have an average particle diameter in a range from 25 to 250 nanometers (nm). The Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a Brunauer-Emmett-Teller (BET) surface area of at least 45 square meters per gram (m²·g⁻¹).

(7) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a BET surface area of at least 50 m²·g⁻¹.

(8) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a BET surface area of at least 55 m²·g⁻¹.

(9) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume of at least 0.10 cubic centimeters per gram (cm³·g⁻¹).

(10) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume of at least 0.15 cm³·g⁻¹.

(11) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume of at least 0.2 cm³·g⁻¹.

(12) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has an average pore diameter of 16.39 nm.

(13) In some embodiments, the spherical metal oxide nanoparticles have an average particle diameter in a range from 50 to 150 nm.

(14) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a mass ratio of the Pd@PdO:CaSiO₃:g-C₃N₄ phases in a range from 0.1-15:0.1-15:0.1-15.

(15) In another exemplary embodiment, a method for making the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material. The method includes autoclaving a solution including calcium nitrate (Ca(NO₃)₂), sodium metasilicate (Na₂SiO₃), and ethanol (C₂H₅OH) at a temperature from 150 to 210 degrees Celsius (° C.) for 1 to 3 hours (h) to form a first reaction mixture. The method further includes dispersing the first reaction mixture in distilled water, filtering, and rinsing with distilled water to form CaSiO₃ and drying the CaSiO₃ at 100 to 140° C. for 1 to 2 h to form a dried CaSiO₃. The method further includes heating urea in a crucible at a temperature from 500 to 700° C. for 0.5 to 2 h to form g-C₃N₄ and heating a reactive solution including palladium chloride (PdCl₂), xylose, and nitric acid (HNO₃) at 100 to 140° C. for 2 to 4 h to form a Pd intermediate. The method further includes calcining the Pd intermediate at a 500 to 600° C. for 3 to 5 h to form Pd@PdO and dispersing the dried CaSiO₃, g-C₃N₄, and Pd@PdO in ethylene glycol monomethyl ether to form a second reaction mixture. The method further includes microwaving the second reaction mixture at 160 to 200° C. and 4 to 6 bar pressure for 0.5 to 2 h to form a first intermediate and dispersing the first intermediate in distilled water, filtering, rinsing with distilled water, and finally, drying at 120 to 180° C. for 1 to 4 h to form the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material.

(16) In some embodiments, the concentration of Ca(NO₃)₂ in the first reaction mixture is in a range from 0.1 to 10 g/L.

- (17) In some embodiments, the concentration of Na_2SiO_3 in the first reaction mixture is in a range from 0.1 to 10 g/L.
- (18) In some embodiments, the concentration of $\text{C}_2\text{H}_5\text{OH}$ in the first reaction mixture is in a range from 400 to 600 milliliters per liter (mL/L).
- (19) In some embodiments, the urea is heated in the crucible at a temperature of 600° C. for 45 minutes (min).
- (20) In some embodiments, the concentration of PdCl_2 in the reactive solution is in a range from 12 to 24 grams per litre (g/L).
- (21) In some embodiments, the concentration of xylose in the reactive solution is in a range from 35 to 55 g/L.
- (22) In some embodiments, the concentration of HNO_3 in the reactive solution is in a range from 80 to 100 ml/L.
- (23) In some embodiments, the Pd intermediate is calcined at 550° C. for 4 h to form Pd@PdO .
- (24) In some embodiments, the second reaction mixture is microwaved at 180° C. and 5 bar pressure.
- (25) In some embodiments, the second reaction mixture is microwaved for 1 hour.
- (26) The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure, and are not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete appreciation of this disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:
- (2) FIG. 1 is an exemplary flowchart for preparing a $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material, according to certain embodiments.
- (3) FIG. 2 shows an X-ray diffraction (XRD) diffractogram of the $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material, according to certain embodiments.
- (4) FIG. 3A shows a transmission electron microscopy (TEM) image of the $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material at 200 nm scale, according to certain embodiments.
- (5) FIG. 3B shows a selected area electron diffraction (SAED) pattern of the $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material, according to certain embodiments.
- (6) FIG. 3C shows an inverse fast Fourier transform (IFFT) measurement of the HRTEM image of the $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material, according to certain embodiments.
- (7) FIG. 3D shows a fast Fourier transform (FFT) measurement of the HRTEM image of the $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material, according to certain embodiments.
- (8) FIG. 4A shows N_2 adsorption-desorption isotherms of the $\text{Pd@PdO/CaSiO}_3\text{-g-C}_3\text{N}_4$ nanocomposite material, according to certain embodiments.

(9) FIG. 4B shows the pore size distribution curves for the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material, plotted using the Barrett-Joyner-Halenda (BJH) method, according to certain embodiments.

DETAILED DESCRIPTION

(10) When describing the present disclosure, the terms used are to be construed in accordance with the following definitions, unless a context dictates otherwise.

(11) Embodiments of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, wherever applicable, in that some, but not all, embodiments of the disclosure are shown.

(12) In the drawings, like reference numerals designate identical or corresponding parts throughout the several views. Further, as used herein, the words 'a', 'an', and the like generally carry a meaning of 'one or more', unless stated otherwise.

(13) Furthermore, the terms 'approximately', 'approximate', 'about', and similar terms generally refer to ranges that include the identified value within a margin of 20%, 10%, or preferably 5%, and any values therebetween.

(14) In this application, a numerical value interval (i.e., a numerical value range) is involved, and, if not specifically stated, an optional numerical value distribution is considered continuous within the numerical value interval, and includes two numerical value endpoints (i.e., minimum and maximum values) of the numerical value range, and each numerical value between the two numerical value endpoint.

(15) As used herein, the term 'room temperature' refers to a temperature range of '25 degrees Celsius ($^{\circ}$ C.) $\pm$ 3 $^{\circ}$ C. in the present disclosure.

(16) As used herein, the term 'nanoparticles' refers to particles having a particle size of 1 nanometer (nm) to 1000 nm within the scope of the present disclosure.

(17) As used herein, the term 'nanocomposite' refers to a composite material that has at least one component with a grain size measured in nm.

(18) As used herein, the term 'nanocomposite materials' refers to multi-phase materials composed of a matrix embedded with nanoscale fillers or components, typically less than 1000 nm in size.

(19) As used herein, the term 'average particle diameter' refers to the mean size of particles in a given sample, typically measured in nanometers (nm), micrometers (μ m), or millimeters (mm), depending on the material.

(20) As used herein, the term 'porosity' refers to a measure of the void or vacant spaces within a material.

(21) As used herein, the term 'pore diameter' refers to an average width or size of the pores (void spaces) within a material, typically measured in nm or angstroms ($\text{\AA}$). It is a useful parameter in characterizing the texture and permeability of porous materials, influencing their adsorption, filtration, or catalytic properties. The pore diameter is often determined using methods such as nitrogen adsorption or mercury intrusion, which provide insights into the material's ability to absorb or interact with molecules of specific sizes.

(22) As used herein, the term 'pore volume' refers to the total volume of void spaces (pores) within a material that is capable of being filled by a gas or liquid. It is typically expressed in cubic centimeters per gram ($\text{cm}^3 \text{g}^{-1}$) and is a useful parameter in characterizing the porous structure of materials, such as adsorbents or catalysts.

(23) As used herein, the term 'amount' refers to the proportion or presence of a particular substance, component, or element within a larger system. It can be quantified in various ways, such as percentage, concentration, or mass. The content of an element refers to its proportion or

concentration within a given material, mixture, or compound. It can be expressed in various units such as atomic percent (at. %), weight percent (wt. %), or mole fraction.

(24) A wt. % of a component, unless specifically stated to the contrary, is based on the total weight of the formulation or composition in which the component is included. For example, if a particular element or component in a composition or article is said to have 5 wt. %, it is understood that this percentage is in relation to a total compositional percentage of 100%.

(25) The present disclosure is intended to include all hydration states of a given compound or formula, unless otherwise noted or when heating a material.

(26) In addition, the present disclosure is intended to include all isotopes of atoms occurring in the present compounds and complexes. Isotopes include those atoms having the same atomic number but different mass numbers. By way of general example, and without limitation, isotopes of hydrogen include deuterium and tritium, and isotopes of carbon include ^{13}C and ^{14}C . Isotopes of oxygen include ^{16}O , ^{17}O , and ^{18}O . Isotopes of nitrogen include ^{14}N and ^{15}N . Isotopes of Ca include ^{40}Ca , ^{42}Ca , ^{43}Ca , ^{44}Ca , and ^{46}Ca . Isotopes of silicon include ^{28}Si , ^{29}Si , and ^{30}Si . Isotopes of hydrogen include ^1H , ^2H , and ^3H . Isotopically-labeled compounds of the disclosure may generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described herein, using an appropriate isotopically-labeled reagent in place of the non-labeled reagent otherwise employed.

(27) Aspects of this disclosure are directed to a $\text{Pd}@\text{PdO}/\text{CaSiO}_3/\text{g-C}_3\text{N}_4$ nanocomposite material fabricated via a multi-step solvothermal precipitation method. The successful integration of palladium (Pd), palladium oxide (PdO), calcium silicate (CaSiO_3), and graphitic carbon nitride ($\text{g-C}_3\text{N}_4$) yields a precisely engineered nanocomposite material achieving uniform distribution, controlled morphology, and multi-component integration. The synergistic effect of multiple components leads to the nanocomposite material being a promising platform for nanotechnology applications.

(28) According to a first aspect of the present disclosure, a $\text{Pd}@\text{PdO}/\text{CaSiO}_3/\text{g-C}_3\text{N}_4$ nanocomposite material (also referred to as the nanocomposite or nanocomposite material) is described. The $\text{Pd}@\text{PdO}/\text{CaSiO}_3/\text{g-C}_3\text{N}_4$ nanocomposite material includes spherical metal oxide nanoparticles, including a $\text{Pd}@\text{PdO}$ phase and a CaSiO_3 phase, dispersed on a matrix of $\text{g-C}_3\text{N}_4$ nanosheets. In some embodiments, PdO exists in the tetragonal phase. In some embodiments, PdO may exist in another crystalline phase, including cubic. In some embodiments, CaSiO_3 exists in the monoclinic phase. In some embodiments, CaSiO_3 may exist in other crystalline phases, including triclinic and orthorhombic crystalline phases. In some embodiments, $\text{g-C}_3\text{N}_4$ has a semi-crystalline phase.

(29) The metal oxide nanoparticles included in the nanocomposite may exist in various morphological shapes, such as nanotubes, nanowires, nanospheres, nanocrystals, nanorectangles, nanotriangles, nanopentagons, nanohexagons, nanoprisms, nanodisks, nanocubes, nanoribbons, nanoblocks, nanobeads, nanotoroids, nanodiscs, nanobarrels, nanogranules, nanowhiskers, nanoflakes, nanofoils, nanopowders, nanoboxes, nanostars, tetrapods, nanobelts, nano-urchins, nanoflowers, etc., and mixtures thereof. In a preferred embodiment, the metal oxide nanoparticles are spherical.

(30) In some embodiments, the spherical metal oxide nanoparticles have an average particle diameter in a range from 10 to 500 nm, preferably 25 to 250 nm, preferably 30 to 245 nm, preferably 35 to 240 nm, preferably 40 to 235 nm, preferably 45 to 230 nm, preferably 50 to 225 nm, preferably 55 to 220 nm, preferably 60 to 215 nm, preferably 65 to 210 nm, preferably 70 to 205 nm, preferably 75 to 200 nm, preferably 80 to 195 nm, preferably 95 to 190 nm, and preferably 100 to 185 nm.

(31) In some embodiments, the spherical metal oxide nanoparticles have an average particle diameter in a range from 50 to 150 nm, preferably 52 to 148 nm, preferably 54 to 146 nm, preferably 56 to 144 nm, preferably 58 to 142 nm, preferably 60 to 140 nm, preferably 62 to 138 nm, preferably 64 to 136 nm, preferably 66 to 134 nm, preferably 68 to 132 nm, preferably 70 to

130 nm, preferably 72 to 128 nm, preferably 74 to 126 nm, preferably 76 to 124 nm, preferably 78 to 122 nm, preferably 80 to 120 nm, preferably 82 to 118 nm, preferably 84 to 116 nm, preferably 86 to 114 nm, preferably 88 to 112 nm, preferably 90 to 110 nm, preferably 92 to 108 nm, preferably 94 to 106 nm, and preferably 96 to 104 nm. In a preferred embodiment, the spherical metal oxide nanoparticles have an average particle diameter of 103 nm.

(32) In one or more embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a mass ratio of the Pd@PdO:CaSiO₃:g-C₃N₄ phases in a range from 0.1-15:0.1-15:0.1-15, preferably 0.25-10:0.25-10:0.25-10, preferably 0.5-7.5:0.5-7.5:0.5-7.5, preferably 0.75-6:0.75-6:0.75-6, preferably 1-5:1-5:1-5.

(33) The Brunauer-Emmett-Teller (BET) hypothesis is the foundation for a useful analysis method for determining the specific surface area of a material. It attempts to explain the physical adsorption of gas molecules on a solid surface. Specific surface area is a property of solids, which is the total surface area of a material per unit of mass, solid or bulk volume, or cross-sectional area. In some embodiments, pore diameter, pore volume, and BET surface area are measured by gas adsorption analysis, preferably N₂ adsorption analysis (e.g., N₂ adsorption isotherms).

(34) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a BET surface area of at least 30 square meters per gram (m²·g⁻¹), preferably at least 45 m²·g⁻¹, preferably at least 46 m²·g⁻¹, preferably at least 47 m²·g⁻¹, preferably at least 48 m²·g⁻¹, preferably at least 49 m²·g⁻¹, preferably at least 50 m²·g⁻¹, preferably at least 51 m²·g⁻¹, preferably at least 52 m²·g⁻¹, preferably at least 53 m²·g⁻¹, preferably at least 54 m²·g⁻¹, preferably at least 55 m²·g⁻¹, preferably at least 56 m²·g⁻¹.

(35) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a BET surface area of at least 50 m²·g⁻¹, preferably at least 50.5 m²·g⁻¹, preferably at least 51.0 m²·g⁻¹, preferably at least 51.5 m²·g⁻¹, preferably at least 52.0 m²·g⁻¹, preferably at least 52.5 m²·g⁻¹, preferably at least 53.0 m²·g⁻¹, preferably at least 53.5 m²·g⁻¹, preferably at least 54.0 m²·g⁻¹, preferably at least 54.5 m²·g⁻¹, preferably at least 55.0 m²·g⁻¹, preferably at least 55.5 m²·g⁻¹, preferably at least 56.0 m²·g⁻¹, and preferably at least 56.5 m²·g⁻¹.

(36) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a BET surface area of at least 55.0 m²·g⁻¹, preferably at least 55.1 m²·g⁻¹, preferably at least 55.2 m²·g⁻¹, preferably at least 55.3 m²·g⁻¹, preferably at least 55.4 m²·g⁻¹, preferably at least 55.5 m²·g⁻¹, preferably at least 55.6 m²·g⁻¹, preferably at least 55.7 m²·g⁻¹, preferably at least 55.8 m²·g⁻¹, preferably at least 55.9 m²·g⁻¹, preferably at least 56.0 m²·g⁻¹, preferably at least 56.1 m²·g⁻¹, preferably at least 56.2 m²·g⁻¹, preferably at least 56.3 m²·g⁻¹, preferably at least 56.4 m²·g⁻¹, preferably at least 56.5 m²·g⁻¹, preferably at least 56.6 m²·g⁻¹, and preferably at least 56.7 m²·g⁻¹. In a preferred embodiment, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a BET surface area equal to 56.8 m²·g⁻¹.

(37) In some embodiments, an average pore distribution of the nanocomposite may include, but is not limited to, crystalline average pore distribution, bimodal, trimodal, multimodal, narrow, broad, and Gaussian. In a preferred embodiment, the average pore distribution of the nanocomposite is unimodal, indicating a single dominant pore size population within the material. This unimodal distribution indicates that the nanocomposites possess a relatively narrow pore size range, resulting in more uniform pore structures and enhancing specific properties such as adsorption capacity, catalytic activity, or surface reactivity.

(38) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has an average pore diameter of 5 to 35 nm, preferably of 10 to 25 nm, preferably 11 to 24

nm, preferably 12 to 23 nm, and preferably 13 to 22 nm, as determined by BJH desorption analysis. In a preferred embodiment, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has an average pore diameter of 16.39 nm.

(39) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume of at least 0.05 cm³.g⁻¹, preferably at least 0.10 cm³.g⁻¹, preferably at least 0.11 cm³.g⁻¹, preferably at least 0.12 cm³.g⁻¹, preferably at least 0.13 cm³.g⁻¹, preferably at least 0.14 cm³.g⁻¹, preferably at least 0.15 cm³.g⁻¹, preferably at least 0.16 cm³.g⁻¹, preferably at least 0.17 cm³.g⁻¹, preferably at least 0.18 cm³.g⁻¹, preferably at least 0.19 cm³.g⁻¹, and preferably at least 0.20 cm³.g⁻¹.

(40) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume of at least 0.150 cm³.g⁻¹, preferably at least 0.155 cm³.g⁻¹, preferably at least 0.160 cm³.g⁻¹, preferably at least 0.165 cm³.g⁻¹, preferably at least 0.170 cm³.g⁻¹, preferably at least 0.175 cm³.g⁻¹, preferably at least 0.180 cm³.g⁻¹, preferably at least 0.185 cm³.g⁻¹, preferably at least 0.190 cm³.g⁻¹, preferably at least 0.195 cm³.g⁻¹, and preferably at least 0.200 cm³.g⁻¹.

(41) In some embodiments, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume of at least 0.200 cm³.g⁻¹, preferably at least 0.201 cm³.g⁻¹, preferably at least 0.202 cm³.g⁻¹, preferably at least 0.203 cm³.g⁻¹, and preferably at least 0.204 cm³.g⁻¹. In a preferred embodiment, the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material has a pore volume equal to about 0.205 cm³.g⁻¹.

(42) FIG. 1 illustrates a flow chart of a method **50** of preparing the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material. The above-mentioned method involves the controlled formation of solid particles from a solution through chemical reactions that lead to supersaturation and subsequent nucleation. The order in which the method **50** is described is not intended to be construed as a limitation, and any number of the described method steps can be combined in any order to implement the method **50**. Additionally, individual steps may be removed or skipped from the method **50** without departing from the spirit and scope of the present disclosure.

(43) At step **52**, the method **50** includes autoclaving a solution including calcium nitrate (Ca(NO₃)₂), sodium metasilicate (Na₂SiO₃), and ethanol (C₂H₅OH) to form a first reaction mixture at a temperature from 120 to 250 degrees Celsius (° C.), preferably 150 to 210 degrees ° C., preferably 155 to 205° C., preferably 160 to 200° C., preferably 165 to 195° C., preferably 170 to 190° C., and preferably 175 to 185° C. for 0.5 to 4.0 hours (h), preferably 1.0 to 3.0 h, preferably 1.1 to 2.9 h, preferably 1.2 to 2.8 h, preferably 1.3 to 2.7 h, preferably 1.4 to 2.6 h, preferably 1.5 to 2.5 h, preferably 1.6 to 2.4 h, preferably 1.7 to 2.3 h, preferably 1.8 to 2.2 h, and preferably 1.9 to 2.1 h. In a preferred embodiment, autoclaving a solution including Ca(NO₃)₂, Na₂SiO₃, and C₂H₅OH to form a first reaction mixture at a temperature of 180° C. for 2 h. This step involves the chemical reaction and dehydration process that are required for converting the precursor materials into the desired solid product. The solution is preferably heated in an autoclave, optionally, other known heating appliances may be used as well.

(44) In some embodiments, the calcium ion source used is Ca(NO₃)₂. In some embodiments, calcium ion source may include but is not limited to calcium chloride, calcium sulfate, calcium carbonate, calcium phosphate, calcium acetate, calcium citrate, calcium lactate, calcium gluconate, calcium formate, calcium oxalate, calcium tartrate, calcium ascorbate, calcium benzoate, calcium propionate, calcium stearate, calcium hydroxide, calcium peroxide, calcium iodate, calcium molybdate, calcium hypochlorite, calcium thiocyanate, calcium chromate, calcium ferrite, calcium bromide, calcium fluoride, calcium sulfide, calcium arsenate, calcium tungstate, calcium borate, calcium perchlorate, calcium hydride, and combinations thereof.

(45) In some embodiments, the concentration of calcium nitrate in the first reaction mixture is in a range from 0.1 to 10 g/L. In some embodiments, the alkali metal silicate is Na_2SiO_3 . In some embodiments, the alkali metal silicate may include but is not limited to potassium silicate, lithium silicate, rubidium silicate, cesium silicate, sodium orthosilicate, potassium orthosilicate, lithium orthosilicate, rubidium orthosilicate, cesium orthosilicate, sodium disilicate, potassium disilicate, lithium disilicate, rubidium disilicate, cesium disilicate, sodium trisilicate, potassium trisilicate, lithium trisilicate, rubidium trisilicate, cesium trisilicate, sodium tetrasilicate, potassium tetrasilicate, lithium tetrasilicate, rubidium tetrasilicate, cesium tetrasilicate, sodium hexasilicate, potassium hexasilicate, lithium hexasilicate, rubidium hexasilicate, cesium hexasilicate, and combinations thereof. In a preferred embodiment, alkali metal silicate is sodium metasilicate.

(46) In some embodiments, the concentration of sodium metasilicate in the first reaction mixture is in a range from 0.1 to 10 g/L.

(47) In some embodiments, the molar ratio of the $\text{Ca}(\text{NO}_3)_2$ to the Na_2SiO_3 may include, but not limited to, 1:6 to 6:1, preferably 1:5 to 5:1, preferably 1:4 to 4:1, preferably 3:1 to 1:3, preferably 1:2 to 2:1, and preferably 1:1. In a preferred embodiment, the molar ratio of the $\text{Ca}(\text{NO}_3)_2$ to the Na_2SiO_3 is 1:1.

(48) In a preferred embodiment, ethanol is used in step **52**, however, other C1-C4 alkanols may be used. In some embodiments, C1-C4 alkanol may include but is not limited to methanol, propanol, isopropanol, butanol, isobutanol, sec-butanol, tert-butanol, 1-propanol, 2-propanol, 1-butanol, 2-butanol, methylpropanol, dimethylpropanol, ethylpropanol, cyclopropanol, fluoromethanol, chloromethanol, bromomethanol, and iodomethanol. The volume-by-volume (v/v) ratio of water to ethanol is in the range of 1:6 to 6:1, preferably 1:5 to 5:1, preferably 1:4 to 4:1, preferably 1:3 to 3:1, preferably 1:2 to 2:1, preferably 1:1. In a preferred embodiment, the v/v ratio of water to ethanol is 1:1.

(49) In some embodiments, the concentration of $\text{C}_2\text{H}_5\text{OH}$ in the first reaction mixture is in a range from 200 to 800 milliliters per litre (mL/L), preferably 400 to 600 mL/L, preferably 420 to 580 mL/L, preferably 440 to 560 mL/L, preferably 460 to 540 mL/L, and preferably 480 to 520 mL/L. In a preferred embodiment, the concentration of $\text{C}_2\text{H}_5\text{OH}$ in the first reaction mixture is 500 mL/L.

(50) At step **54**, the method **50** includes dispersing the first reaction mixture in distilled water, filtering, and rinsing with distilled water to form CaSiO_3 . In some embodiments, the solid particles of the first reaction mixture are separated by distinct methods, which may include, but are not limited to, gravity filtration, centrifugation, hot filtration, cold filtration, granular media filtration, mechanical filtration, multilayer filtration, and vacuum filtration. In a preferred embodiment, solid crystalline nanocomposite from the heated dispersion is separated by vacuum filtration using a Buchner filtration system. The water may be tap water, distilled water, bi-distilled water, deionized water, deionized distilled water, reverse osmosis water, and/or some other water. In a preferred embodiment, the water for rinsing in the Buchner filtration system is distilled water.

(51) At step **56**, the method **50** includes drying the CaSiO_3 to form a dried CaSiO_3 at 80 to 200° C., preferably 100 to 140° C., preferably 105 to 135° C., preferably 110 to 130° C., and preferably 115 to 125° C. for 0.25 to 5 h, preferably 0.5 to 3 h, preferably 0.6 to 2.9 h, preferably 0.7 to 2.8 h, preferably 0.8 to 2.7 h, and preferably 0.9 to 2.6 h. In some embodiments, drying the CaSiO_3 is performed for 1 to 2 h. In a preferred embodiment, drying the CaSiO_3 to form a dried CaSiO_3 is performed at 120° C. for 1 h to evaporate the solvents completely. The first reaction mixture may be heated using any heating appliances such as ovens, microwaves, autoclaves, hot plates, heating mantles and tapes, oil baths, salt baths, sand baths, air baths, hot-tube furnaces, and hot-air guns.

(52) At step **58**, the method **50** includes heating urea in a crucible to form g- C_3N_4 at a temperature from 400 to 800° C., preferably 500 to 700° C., preferably 520 to 680° C., preferably 540 to 660° C., preferably 560 to 640° C., and preferably 580 to 620° C. for a period of 15 minutes to 90 minutes, preferably 30 minutes to 60 minutes, preferably 40 minutes to 50 minutes, preferably about 45 minutes. Heating is done in a closed vessel at step **58** to maintain a controlled atmosphere, preventing the escape of volatile components and leading to the formation of g-

C.sub.3N.sub.4 under the required conditions of pressure and temperature. In a preferred embodiment, the urea is heated to a temperature of about 600° C. for a period of about 45 minutes.

(53) At step **60**, the method **50** includes heating a reactive solution including palladium chloride (PdCl.sub.2), xylose, and nitric acid (HNO.sub.3) to form a palladium intermediate at 80 to 180° C., preferably 100 to 140° C., preferably 105 to 135° C., preferably 110 to 130° C., and preferably 115 to 125° C. for 1 to 5 h, preferably 2 to 4 h, preferably 2.2 to 3.8 h, preferably 2.4 to 3.6 h, preferably 2.6 to 3.4 h, and preferably 2.8 to 3.2 h. In a preferred embodiment, heating a reactive solution including palladium chloride (PdCl.sub.2), xylose, and HNO.sub.3 to form a palladium intermediate is performed at 120° C. for 3 h.

(54) In some embodiments, the palladium salt used is palladium chloride (PdCl.sub.2). In some embodiments, palladium ion source may include but is not limited to palladium chloride, palladium sulfate, palladium carbonate, palladium phosphate, palladium acetate, palladium citrate, palladium lactate, palladium gluconate, palladium formate, palladium oxalate, palladium tartrate, palladium ascorbate, palladium benzoate, palladium propionate, palladium stearate, palladium hydroxide, palladium (IV) peroxide, palladium iodate, palladium molybdate, palladium hypochlorite, palladium thiocyanate, palladium chromate, palladium ferrite, palladium bromide, palladium fluoride, palladium sulfide, palladium arsenate, palladium tungstate, palladium borate, palladium perchlorate, palladium hydride, and combinations thereof.

(55) In some embodiments, the reducing sugar used is xylose. In some embodiments, the reducing sugar is a monosaccharide selected from the group consisting of trioses, tetroses, pentoses, hexoses, heptoses, and combinations thereof. In some embodiments, the reducing sugar may include but is not limited to the erythrose, threose, erythrulose, ribose, arabinose, xylose, lyxose, ribulose, arabulose, xylulose, lyxulose, glucose, mannose, galactose, allose, altrose, talose, gulose, idose, fructose, psicose, sorbose, tagatose, sedoheptulose, and combinations thereof.

(56) In some embodiments, sulfuric acid (H.sub.2SO.sub.4), being a strong acid, may be used in the reaction at step **60** as it may effectively break down palladium oxychloride octahydrate and facilitate the formation of a palladium intermediate. Hydrochloric acid (HCl) is capable of dissolving metal chlorides and promoting the reaction. Phosphoric acid (H.sub.3PO.sub.4) may also be used, including if a more controlled reaction or specific palladium phases are desired. The choice of acid depends on the desired reactivity and the final properties of the palladium-based material. In a preferred embodiment, the acid used in the reaction to form palladium intermediate is nitric acid (HNO.sub.3).

(57) In some embodiments, the concentration of PdCl.sub.2 in reactive solution is in a range from 5 to 35 g/L, preferably 12 to 24 g/L, preferably 13 to 23 g/L, preferably 14 to 22 g/L, and preferably 15 to 21 g/L. In a preferred embodiment, the concentration of PdCl.sub.2 in a reactive solution is 18 g/L.

(58) In some embodiments, the concentration of xylose in the reactive solution is in a range from 25 to 65 g/L, preferably 35 to 55 g/L, preferably 36 to 54 g/L, preferably 37 to 53 g/L, preferably 38 to 52 g/L, and preferably 39 to 51 g/L. In a preferred embodiment, the concentration of xylose in the reactive solution is 50 g/L.

(59) In some embodiments, the concentration of HNO.sub.3 in the reactive solution is in a range from 60 to 120 mL/L, preferably 80 to 100 mL/L, preferably 82 to 98 mL/L, preferably 84 to 96 mL/L, and preferably 86 to 94 mL/L, and preferably 88 to 92 mL/L. In a preferred embodiment, the concentration of HNO.sub.3 in the reactive solution is 90 mL/L.

(60) At step **62**, the method **50** includes calcining the palladium intermediate to form Pd@PdO at 400 to 700° C., preferably 500 to 600° C., preferably 510 to 590° C., preferably 520 to 580° C., preferably 530 to 570° C., and preferably 540 to 560° C. for 2 to 6 h, preferably 3 to 5 h, preferably 3.2 to 4.8 h, preferably 3.4 to 4.6 h, preferably 3.6 to 4.4 h, and preferably 3.8 to 4.2 h. In a preferred embodiment, calcining the palladium intermediate to form Pd@PdO is performed at 550° C. for 4 h. The calcination is carried out by heating it to a high temperature under a restricted supply of ambient oxygen. This is performed to remove impurities or volatile substances and to incur thermal decomposition. Typically, the calcination is carried out in a furnace, preferably

equipped with a temperature control system. This process leads to the formation of a Pd@PdO. The high temperature accelerates the reaction, promoting the required chemical transformations.

(61) At step **64**, the method **50** includes dispersing the dried CaSiO₃, g-C₃N₄, and Pd@PdO in ethylene glycol monomethyl ether to form a second reaction mixture. In some embodiments, ethylene glycol monomethyl ether is used, however other polar protic solvents may be used including, but not limited to, water, methanol, ethanol, propanol, isopropanol, butanol, isobutanol, sec-butanol, tert-butanol, pentanol, hexanol, heptanol, octanol, nonanol, decanol, formic acid, acetic acid, propionic acid, butyric acid, lactic acid, glycolic acid, glycerol, ethylene glycol, propylene glycol, diethylene glycol, triethylene glycol, 1,2-butanediol, 1,3-butanediol, 1,2-pentanediol, 1,3-propanediol and combinations thereof. In a preferred embodiment, the polar protic solvent used is ethylene glycol monomethyl ether.

(62) At step **66**, the method **50** includes microwaving the second reaction mixture to form a first intermediate at 120 to 240° C., preferably 160 to 200° C., preferably 165 to 195° C., preferably 170 to 190° C., and preferably 175 to 185° C., and 2 to 8 bar, preferably 4 to 6 bar, preferably 4.1 to 5.9 bar, preferably 4.2 to 5.8 bar, preferably 4.3 to 5.7 bar, preferably 4.4 to 5.6 bar, preferably 4.5 to 5.5 bar, preferably 4.6 to 5.4, preferably 4.7 to 5.3 bar, preferably 4.8 to 5.2 bar, and preferably 4.9 to 5.1 bar pressure for 0.25 to 4 h, preferably 0.5 to 2 h, preferably 0.6 to 1.9 h, preferably 0.7 to 1.8 h, preferably 0.8 to 1.7 h, preferably 0.7 to 1.8 h, preferably 0.8 to 1.7 h, and preferably 0.9 to 1.6 h. In a preferred embodiment, microwaving the second reaction mixture to form a first intermediate at 180° C. and 5.0 bar pressure for 1.0 h. The solution is preferably heated in a microwave; optionally, other known heating appliances may be used as well.

(63) At step **68**, the method **50** includes dispersing the first intermediate in distilled water, filtering, rinsing with distilled water, and drying at 80 to 220° C., preferably 120 to 180° C., preferably 125 to 175° C., preferably 130 to 170° C., preferably 135 to 165° C., preferably 140 to 160° C., and preferably 145 to 155° C. for 0.5 to 6 h, preferably 1 to 4 h, preferably 1.2 to 3.8 h, preferably 1.4 to 3.6 h, preferably 1.6 to 3.4 h, and preferably 1.8 to 3.2 h to form the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material. In a preferred embodiment, after dispersing the first intermediate in distilled water, filtering, and rinsing with distilled water, drying is performed at 150° C. for 2 h to form the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material.

(64) In some embodiments, the solid nanocomposite material is separated by distinct methods, which may include, but are not limited to, gravity filtration, centrifugation, hot filtration, cold filtration, granular media filtration, mechanical filtration, multilayer filtration, and vacuum filtration. In a preferred embodiment, the solid nanocomposite material is separated by vacuum filtration using a Buchner filtration system. The water may be tap water, distilled water, bidistilled water, deionized water, deionized distilled water, reverse osmosis water, and/or some other water. In a preferred embodiment, the water is distilled water for rinsing in the Buchner filtration system.

EXAMPLES

(65) The following examples demonstrate a Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material (herein referred to as the nanocomposite or nanocomposite material). The examples are provided solely for illustration and are not to be construed as limitations of the present disclosure, as many variations thereof are possible without departing from the spirit and scope of the present disclosure.

Example 1: Fabricating the Pd@PdO/CaSiO₃@g-C₃N₄

(66) Equal moles of calcium nitrate (0.5 g) and sodium metasilicate (0.37 g) were dispersed in 100 milliliters (mL) of ethanol:water (1:1) in a 150 mL glass beaker and sonicated for 15 minutes. The mixture was transferred to a 200 mL autoclave and then placed in an oven operated at 180° C. for 2 hours. The product was dispersed in 500 ml distilled water with an ultrasonic bath for 10 minutes, filtered via a Buchner system, rinsed with distilled water, and dried at 120° C. for 1 hour.

(67) About 30 grams (g) of urea (NH₂CONH₂) was placed in a 250 milliliters (mL) porcelain crucible and covered with its porcelain cover, then the crucible and cover were wrapped with three layers of aluminum foil to reduce the NH₂CONH₂ loss to sublimation. The crucible was heated via a furnace set at 600° C. for 45 min.

(68) About 2 g of palladium chloride (PdCl₂) and 5 g of xylose were placed in a 500 ml beaker. 100 ml distilled water was added to the mixture and heated till a clear solution was obtained. 10 mL of concentrated nitric acid (HNO₃) was added to the mixture, which was then heated till the carbonization of C₅H₁₀O₅. The mixture was placed in an oven set at 120° C. for 3 h; the black product was milled in a mortar, placed in a 150 mL porcelain dish, and calcined at 550° C. for 4 h.

(69) An equal amount of calcium silicate (CaSiO₃), graphitic carbon nitride (g-C₃N₄), and palladium@palladium oxide (Pd@PdO) (0.5 grams each) were transferred to a mono wave-200 vial (G30) and dispersed in 20 mL ethylene glycol monomethyl ether via an ultrasonic bath for 30 minutes. The vial was closed with its Teflon cover and placed in the Anton-Baar Monowave-200 operated at 180° C. and 5 bar pressure for 1 h. The product was dispersed in 1 liter (L) distilled water with an ultrasonic bath for 30 minutes, filtered via a Buchner system, rinsed with distilled water, and dried at 150° C. for 2 h.

(70) The crystallinity and phases identification present in Pd@PdO/CaSiO₃@g-C₃N₄ catalyst were analyzed by X-ray diffraction (XRD), and the results are given in FIG. 2. The intense peaks and high-intensity values indicated the highly crystalline nature of the nanocomposite. In addition, the peak broadening indicates the small particle size of the prepared nanocomposite. Examination of the diffraction patterns with the standard joint committee on powder diffraction standards (JCPDS) cards reveals the presence of metallic Pd, PdO, CaSiO₃, and g-C₃N₄. The diffraction peaks for PdO phase were observed at 2θ values of 33.3°, 34.0°, 42.2°, 54.9°, and 60.5°, corresponding to (002), (101), (110), (112), and (103) planes, respectively, of the tetragonal phase of the PdO phase (JCPDS number (no.) 00-041-1107). Strong diffractions related to metallic Pd (JCPDS no. 00-046-1043) were observed at 2θ values of 40.3°, 46.6°, and 68.4°, corresponding to (111), (200), and (220) planes, respectively. The diffraction peaks for CaSiO₃ were observed at 2θ values of 27.0°, 29.4°, 39.4°, and 48.8°, corresponding to (-202), (-221), (-521), and (-810) planes, respectively, of the monoclinic phase of CaSiO₃ (JCPDS no. 00-043-1460). The diffractions related to g-C₃N₄ were observed at 22.9, 26.7, and 47.1° (COD No. 1534042 and JCPDS no. 01-087-1526). The weak diffractions of both CaSiO₃ and g-C₃N₄ phases demonstrated their semi-crystalline nature, and strong diffractions of both PdO and Pd phases demonstrated their high crystalline nature. No other phases were observed and indicating the successful fabrication of Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material.

(71) A transmission electron microscopy (TEM) image of Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite is represented in FIG. 3A. The TEM image represents a two-dimensional porous structure constructed with curled and wrinkled nanosheets and platelets of the g-C₃N₄. The TEM image also shows a good dispersion of large spherical aggregates of metal oxide nanoparticles with a diameter of 103 nanometers (nm) on nanosheets of g-C₃N₄. The corresponding selected area electron diffraction (SAED) pattern reveals diffraction spots with interplanar spacing of 0.295 nm, 0.220 nm, 0.162, and 0.132 nm due to (CaSiO₃: 221, PdO: 100), (CaSiO₃: 322), (PdO: 112, CaSiO₃: -142), and (PdO: 202, CaSiO₃: 823), diffraction planes (FIG. 3B). The inverse fast Fourier transform (IFFT) and fast Fourier transform (FFT) measurements show a d value of 0.23 nm given to the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite, which correspond to the lattice spacing of (004) indicating the development of CaSiO₃, respectively (FIGS. 3C-3D).

(72) FIG. 4A displays the nitrogen adsorption-desorption isotherms of the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite. The nitrogen sorption isotherm of the nanocomposite belongs to a type IV hysteresis loop, indicating the formation of mesoporous structures. However, shifting the loop to a relatively higher pressure (relative pressure (P/P₀)=0.70-1) indicates the presence of wide mesopores, which may result from the deposition of metal oxide particles in the wide pores of g-C₃N₄. Furthermore, the Brunauer-Emmett-Teller (BET) surface area of the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material was calculated to be 56.8 square meters per gram (m² g⁻¹). The high specific surface area reflects the good dispersion of these metal oxide nanoparticles on g-C₃N₄. Moreover, the pore size distribution curves, plotted using the Barrett-Joyner-Halenda (BJH) method for the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material (FIG. 4B), exhibited an unimodal distribution with average pore diameters maximized at 16.39 nm and pore volume of 0.205 cubic centimeters per gram (cm³ g⁻¹). All the isotherms

belong to the H.sub.3 category of pores, which do not exhibit limiting adsorption at high P/P.sub.0, and arise due to aggregation of plate-like particles, giving rise to slit-shaped pores. This indicates that the assembly of Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite produced a mesoporous structure.

(73) Numerous modifications and variations of the present disclosure are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the disclosure may be practiced otherwise than as specifically described herein.

Claims

1. A Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material, including: spherical metal oxide nanoparticles including a palladium@palladium oxide (Pd@PdO) phase and a calcium silicate (CaSiO.sub.3) phase dispersed on a matrix of graphitic carbon nitride (g-C.sub.3N.sub.4) nanosheets, wherein the spherical metal oxide nanoparticles have an average particle diameter in a range from 25 to 250 nanometers (nm), and wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a Brunauer-Emmett-Teller (BET) surface area of at least 45 square meters per gram (m.sup.2.Math.g.sup.-1).
2. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 1, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a BET surface area of at least 50 m.sup.2.Math.g.sup.-1.
3. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 2, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a BET surface area of at least 55 m.sup.2.Math.g.sup.-1.
4. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 1, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a pore volume of at least 0.10 cubic centimeters per gram (cm.sup.3.Math.g.sup.-1).
5. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 4, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a pore volume of at least 0.15 cm.sup.3.Math.g.sup.-1.
6. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 5, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a pore volume of at least 0.2 cm.sup.3.Math.g.sup.-1.
7. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 1, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has an average pore diameter of 16.39 nm.
8. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 1, wherein the spherical metal oxide nanoparticles have an average particle diameter in a range from 50 to 150 nm.
9. The Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 8, wherein the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material has a mass ratio of the Pd@PdO:CaSiO.sub.3:g-C.sub.3N.sub.4 phases in a range from 0.1-15:0.1-15:0.1-15.
10. A method for making the Pd@PdO/CaSiO.sub.3@g-C.sub.3N.sub.4 nanocomposite material of claim 1, including: autoclaving a solution including calcium nitrate (Ca(NO.sub.3).sub.2), sodium metasilicate (Na.sub.2SiO.sub.3), and ethanol (C.sub.2H.sub.5OH) at a temperature from 150 to 210 degrees Celsius (° C.) for 1 to 3 hours (h) to form a first reaction mixture; dispersing the first reaction mixture in distilled water, filtering, and rinsing with distilled water to form CaSiO.sub.3; drying the CaSiO.sub.3; at 100 to 140° C. for 1 to 2 h to form a dried CaSiO.sub.3; heating urea in a crucible at a temperature from 500 to 700° C. for 0.5 to 2 h to form g-C.sub.3N.sub.4; heating a reactive solution comprising palladium chloride (PdCl.sub.2), xylose, and nitric acid (HNO.sub.3) at 100 to 140° C. for 2 to 4 h to form a Pd intermediate; calcining the Pd intermediate at a 500 to 600° C. for 3 to 5 h to form Pd@PdO; dispersing the dried CaSiO.sub.3, g-C.sub.3N.sub.4, and Pd@PdO in ethylene glycol monomethyl ether to form a second reaction mixture; microwaving the

second reaction mixture at 160 to 200° C. and 4 to 6 bar pressure for 0.5 to 2 h to form a first intermediate; and dispersing the first intermediate in distilled water, filtering, rinsing with distilled water, and drying at 120 to 180° C. for 1 to 4 h to form the Pd@PdO/CaSiO₃@g-C₃N₄ nanocomposite material.

- 11.** The method of claim 10, wherein the concentration of Ca(NO₃)₂ in the first reaction mixture is in a range from 0.1 to 10 g/L.
- 12.** The method of claim 10, wherein the concentration of Na₂SiO₃ in the first reaction mixture is in a range from 0.1 to 10 g/L.
- 13.** The method of claim 10, wherein the concentration of ethanol in the first reaction mixture is in a range from 400 to 600 milliliters per litre (mL/L).
- 14.** The method of claim 10, wherein the urea is heated in the crucible at a temperature of 600° C. for 45 minutes (min).
- 15.** The method of claim 10, wherein the concentration of PdCl₂ in the reactive solution is in a range from 12 to 24 grams per litre (g/L).
- 16.** The method of claim 10, wherein the concentration of xylose in the reactive solution is in a range from 35 to 55 g/L.
- 17.** The method of claim 10, wherein the concentration of HNO₃ in the reactive solution is in a range from 80 to 100 mL/L.
- 18.** The method of claim 10, wherein the palladium intermediate is calcined at 550° C. for 4 h to form Pd@PdO.
- 19.** The method of claim 10, wherein the second reaction mixture is microwaved at 180° C. and 5 bar pressure.
- 20.** The method of claim 10, wherein the second reaction mixture is microwaved for 1 h.